

PERFORMANCE TESTING PHASE

Project Performance Testing

Date	01 JULY 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Model Performance Testing

S. No	Parameter	Screenshot / Values
1	Application Loading	The Smart Connect AI application was successfully launched using Streamlit. All UI components including the sidebar, Event Description, Knowledge Assistant, Save Session, Feedback, and Project Summary loaded correctly without errors.
2	Backend API Testing	FastAPI server was tested using Uvicorn. All REST API endpoints (/generate, /factcheck, /save-history, /feedback) responded successfully with valid JSON outputs.
3	Conversation Starter Generation	The Event Analyzer successfully analysed the networking event description and generated AI-powered conversation starters within a few seconds.
4	Knowledge Assistant Testing	The Wikipedia API successfully returned verified summaries for user-entered topics. The retrieved information was displayed correctly in the Streamlit interface.
5	Session History Testing	Networking session details were successfully stored in history.json after clicking the Save Session button.
6	Feedback Testing	User feedback was successfully stored in feedback.json, and confirmation messages were displayed.

Performance Evaluation

1. Application Loading

Result

- Streamlit UI loaded successfully.
- Sidebar displayed correctly.
- Buttons responded without delay.

- Responsive layout verified.
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2. Backend API Performance

Result

The following API endpoints were tested successfully:

Endpoint	Status
/	Working
/generate	Working
/factcheck	Working
/save-history	Working
/feedback	Working

Average API response time was less than **2–3 seconds** under normal conditions.

3. Conversation Starter Generation

Input

Tesla AI Conference

Output

Example conversation starter:

"What internship opportunities are available at Tesla?"

Status: Successful

4. Knowledge Assistant

Input

Artificial Intelligence

Output

Verified summary retrieved successfully from Wikipedia.

Status: Successful

5. JSON Storage Testing

history.json

Successfully stores:

- Event Description
- Generated Topics
- Timestamp

feedback.json

Successfully stores:

- User Feedback
 - Timestamp
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6. User Interface Testing

The following UI components were tested:

- Sidebar
- Event Description
- Generate Smart Starters
- Knowledge Assistant
- Save Session
- Feedback Form
- Project Summary
- Footer

All components worked correctly.

Functional Testing Results

Feature	Status
Event Description	✓ Passed
Conversation Generator	✓ Passed
Knowledge Assistant	✓ Passed
Wikipedia Integration	✓ Passed

Feature	Status
Save Session	✓ Passed
Feedback Submission	✓ Passed
FastAPI Communication	✓ Passed
Streamlit UI	✓ Passed
JSON Storage	✓ Passed

Technologies Tested

Component	Result
Python	Working
FastAPI	Working
Streamlit	Working
Requests Library	Working
Wikipedia API	Working
JSON Storage	Working
Uvicorn Server	Working

Dashboard / User Interface Testing

The Smart Connect AI dashboard was tested for:

- Layout consistency
- Navigation
- Button functionality
- User interaction
- API connectivity
- Data display

All interface elements responded correctly.

Overall Performance

Parameter	Status
System Stability	Excellent
API Response	Fast
User Interface	Responsive
Data Storage	Successful
Error Handling	Good
Overall Performance	Excellent

Conclusion

The **Personalized Networking Assistant (Smart Connect AI)** successfully passed all functional and performance tests. The Streamlit frontend communicated effectively with the FastAPI backend, AI-generated conversation starters were produced accurately, the Wikipedia API provided reliable knowledge verification, and networking sessions and feedback were stored successfully in JSON files. The application demonstrated stable performance, quick response times, and a responsive user interface, making it suitable for real-world networking preparation and professional event assistance.